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Developing a Deep Learning-Based Computer Vision Framework for Automated Pre- and Post-Rental Passenger Car Damage Inspection

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ABSTRACT

The manual, time-consuming and subjective inspection process for passenger cars in the car rental industry leads to customer disputes, operational inefficiencies and financial losses. In this study, we propose a new solution to the problem by developing and evaluating a robust computer vision framework based on deep learning to automate the detection of external passenger car damage. The project will follow a Design Science Research methodology to develop a functional software artefact. This will include a detailed literature review on deep learning architectures, with particular interest in Convolutional Neural Networks (CNNs), YOLO family of object detection models and transfer learning approaches, and the curation and augmentation of a specialized image dataset. The real-world passenger car inspection images will be collected from DriveTime Rentals PVT LTD, a Sri Lankan car rental operator, and supplemented with publicly available datasets, to make the training data representative of real rental lot conditions. A state-of-the-art YOLOv8 deep learning model fine-tuned using transfer learning from COCO-pretrained weights will be developed for accurate identification, localization and classification of passenger car damages including scratches, dents, cracks and broken components. Standard metrics such as the mean Average Precision (mAP) and precision, recall and F1-score will be calculated for a comprehensive evaluation of model performance. The trained model will be deployed as a Python FastAPI microservice and integrated with a .NET 8 Web API backend which will be consumed by an Angular 17 inspection dashboard allowing rental staff to do real-time annotated pre- and post-rental comparison. This work tackles important open questions in the research community such as computational efficiency, model generalization and data scarcity. It also takes a hard look at the ethics of algorithmic bias, data privacy and automation. The successful completion of this project will provide a proven foundation for a commercial solution that has the potential to significantly increase operational throughput, reduce costs, improve customer satisfaction and create the conditions for the development of fully contactless rental experiences.

Keywords: Deep Learning, Object Detection, YOLOv8, Vehicle Damage Detection, Computer Vision, Design Science Research, Transfer Learning, Full-Stack Software Architecture, Enterprise AI Integration, Explainable AI.

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LIST OF ABBREVIATIONS

Abbreviation	Name
CNNs	Convolutional Neural Network
YOLO	You Only Look Once
ONNX	Open Neural Network Exchange
DSR	Design Science Research
DL	Deep Learning

1 PROJECT INITIATION

1.1 Introduction

The global car rental industry is expected to be worth USD 92 billion in 2023 and millions of vehicle transactions are made every year. However, human inspections at vehicle check-in and check-out remain laborious, uneven, and error-prone (IBISWorld, 2023). For walk-around assessments, staff time per vehicle can be up to fifteen minutes, documentation of condition is inconsistent, and photos rarely create a reliable baseline of before-and-after. This translates into high volumes of open damage disputes, loss of revenue and increasing customer dissatisfaction.

This dissertation describes the design, development, and evaluation of an automated car damage inspection framework utilizing computer vision (CV). We will develop a deep learning model to detect and classify surface damage such as scratches, dents, cracks, broken components from inspection photographs. The CV model will be deployed to a .NET 8 Web API backend and consumed by an Angular 17 inspection dashboard, giving rental staff real-time annotated damage reports and side-by-side pre- and post-rental comparison.

The study is anchored in Design Science Research (Hevner et al., 2004) and contributes to the body of knowledge in academia and industry. It builds on the candidate's professional expertise in .NET and Angular, extending capability into applied AI and computer vision directly supporting career development toward software architecture and AI-integrated systems design.

1.2 Research Background

The global car rental industry is valued at about USD 92 billion in 2023 (IBISWorld, 2023). Accurate vehicle condition assessment at every check-in and check-out cycle is crucial. At present, it's a manual process. An employee walks around the vehicle, checks the bodywork, and records any damage, usually on paper or a simple digital form, and takes photos on a mobile device. This is said to take up to fifteen minutes per vehicle and is very dependent on the individual inspector's keenness, lighting conditions and previous experience. Disputes over damage responsibility are common because the documentation that results is inconsistent and rarely provides a reliable, comparable baseline between the pre-rental and post-rental condition of a vehicle, with European operators reporting an average resolution cost of £1,200 per unresolved case (Deloitte, 2022).

Recent advances in deep learning based computer vision, and in particular single-stage object detection architectures such as the YOLO (You Only Look Once) family, have made it now technically feasible to automate the detection, localisation and classification of common types of exterior vehicle damage such as scratches, dents, cracks and broken components (Redmon et al., 2016; Jocher et al., 2023). But as discussed in Section 2, most of the work published so far looks at the computer vision problem in isolation, evaluated on benchmark datasets, and not how such a model would be deployed, operated and consumed inside a real enterprise software stack that a rental operator could use on a daily basis.

1.3 Research Problem

Manual inspection of the exterior is, despite its centrality to the fleet operations, fundamentally a perception task performed under time pressure. Human visual perception is well documented to be fallible under exactly these conditions . A fifteen-minute walk-around, often at the end of a long shift, in variable outdoor lighting, on a vehicle finish that may already be dirty, scratched from prior use, or simply unfamiliar, demands that an inspector notice, correctly classify and consistently document subtle surface defects, hairline scratches, shallow dents, small cracks, against a constantly changing baseline. The inspector has no objective ground truth at the point of decision. Judgement is exercised from memory and experience alone. This makes the task considerably harder than it first appears, and the error rate is less a matter of individual carelessness than an inherent limitation of unaided human visual assessment repeated at volume: fatigue, inattentional blindness, inconsistent lighting and inter-inspector variation in what counts as ‘damage’ all degrade reliability in ways that are difficult to train away. The practical consequence is that manual inspection routinely results in missed defects, inconsistent severity judgments, and incomplete documentation precisely the failure modes that surface later as disputed, unresolved damage claims between operator and customer.

The research problem addressed by this dissertation is therefore twofold, and both elements follow directly from this underlying difficulty of the manual task. First, there is a technical sub-problem: existing public datasets for vehicle damage detection are not representative of the specific conditions found on a rental lot, such as consistent overhead lighting, narrow camera-angle variation, and a limited and predictable range of fleet vehicle types, which constrains the accuracy of models trained purely on public data when applied to a real operator's environment. An automated system intended to replace or augment human judgement is only as trustworthy as the data it has learned from, so this gap must be closed before any claim of improved reliability over manual inspection can be substantiated. The second is a systems-integration sub-problem. No published study has looked at a full end-to-end pipeline from a fine-tuned deep learning model to a usable enterprise interface, including record management, authentication, evidentiary storage and dispute-support reporting that a non-technical rental staff member could operate without specialist knowledge. If a model with high benchmark accuracy cannot be reliably operated, quickly and consistently, by the same time-pressured staff whose unaided judgement it is meant to support, it will offer no practical reduction in human error.

1.4 Research Objective

The dissertation has six aims:

1. Critically review literature on deep learning architectures for automotive damage detection, transfer learning strategies and full-stack AI deployment patterns published from 2015 to the present.
2. Collect and curate a labelled training dataset combining the CarDD and Roboflow Vehicle Damage Detection public datasets with real world inspection images obtained from DriveTime Rentals PVT LTD.
3. Design, train and fine-tune a YOLOv8 model for multi-class damage detection and benchmark it against a YOLOv5 baseline.
4. Build a .NET 8 Web API that orchestrates inference, manages inspection records and stores evidentiary images securely.

5. Build an Angular 17 dashboard enabling staff to run inspections and generate pre/post comparison reports.
6. Assess the whole system's usability, inspection time savings, reaction latency, and detection accuracy.

1.5 Research Question

What is the design, development and evaluation of a deep learning-based computer vision framework for the automation of pre and post rental exterior damage inspection of passenger cars in a full stack .NET 8 and Angular 17 enterprise system?

This main question is answered by the following sub-questions

1. What deep learning model architecture and transfer learning strategy achieve highest accuracy (mAP, precision, recall) for exterior damage detection and classification such as scratches, dents, cracks, broken parts on passenger cars in rental lot conditions?
2. How well can real-world passenger car inspection images from DriveTime Rentals PVT LTD, along with public datasets, be used to train a domain-specific YOLOv8 model for rental damage detection?
3. How does the incorporation of the deep learning model in the .NET 8 Web API and Angular 17 dashboard reduce passenger car inspection time and improve inspection consistency compared to the current manual process?
4. How reliable is the system in generating structured evidentiary documentation of pre- and post-rental passenger car conditions to support fair dispute resolution in car rental operations?

1.6 Significance of the Study

This study is important in two respects. In an academic sense, it bridges a gap identified in the literature between computer vision research assessing detection accuracy in isolation and information systems research that has not yet explored the practical integration of such models into enterprise architectures (Reddy and Vishnu, 2022). In practice, the resulting prototype provides a proof-of-concept route for car rental operators, and DriveTime Rentals PVT LTD in particular, to faster, more consistent and more defensible vehicle condition documentation, with the potential to reduce costs of disputes and improve customer experience.

1.7 Scope and Limitations

The research is exclusively aimed at the detection of passenger car exterior damage from static photos captured within regular rental inspection processes. Vehicle classes that are not passenger cars, such as commercial vehicles, motorcycles and vans, are out of scope. Does not cover interior damage, mechanical faults or video inspection. The system will be developed and tested as a proof-of-concept prototype, not a production deployment. The user study will be conducted on a small sample of ten volunteers and may not be representative of the diversity of rental staff populations. Model performance can vary with vehicle colour, lighting conditions and damage severities that are poorly represented in the training data.

2 LITERATURE REVIEW

This chapter discusses the literature that underpins the dissertation along three related strands: the development of deep learning-based object detection and its application to vehicle damage assessment; the systems-engineering literature on deploying computer vision models in enterprise software; and the literature on explainability, bias and research ethics relevant to automated inspection. This review is concluded by synthesising these strands into a conceptual framework that motivates the methodology adopted in Chapter 4.

2.1 Industry Background and Vehicle Inspection Problem

Manual vehicle inspection remains the dominant practice throughout the car rental sector despite its well documented inefficiencies. IBISWorld (2023) estimates the global car rental industry to be worth USD 92 billion and processes millions of rental transactions annually, with each transaction requiring a minimum of two manual condition checks. Deloitte (2022) estimates a monetary value to the operational cost of this manual process in relation to dispute resolution, with an average of £1,200 per unresolved damage dispute for European operators, driven mainly by the lack of an objective, time-stamped, comparable record of vehicle condition before and after rental. This provides the practical motivation for the dissertation: a moderate increase in inspection consistency and speed could lead to material operational savings, a justification consistent with the design science principle that research should address a problem of genuine relevance to practice (Hevner et al., 2004).

The inefficiency of manual inspection is not unique to vehicle rental; similar challenges have been documented in adjacent inspection-heavy domains. For example, Nath et al. (2020) study the manual verification of personal protective equipment on construction sites and show that human verification is slow, subjective and hard to audit, which motivates their use of deep learning to perform automated, real-time compliance checking. The similarities between a construction site inspection and a rental vehicle inspection both require a trained, time-pressured human inspector to visually verify a physical asset against a known standard are conducive to the transfer of computer vision-based automation approaches to the rental damage inspection context, as targeted by this dissertation.

2.2 Evaluation of Deep Learning for Object Detection

Convolutional Neural Networks (CNNs) are the foundation for modern computer vision . Deep residual learning was introduced by He et al. (2016) which showed that very deep CNN architectures could be trained without the degradation problem that previously limited network depth, by using residual (skip) connections. This architectural innovation underlies many of the backbone networks used in contemporary object detectors, including those in the YOLO family discussed below, and remains foundational to the feature-extraction stages of the YOLOv8 architecture selected for this dissertation.

When applied to surface damage detection, Cha et al. (2017) showed that a CNN-based classifier could detect structural cracks in concrete with an accuracy comparable to that of a trained human inspector. This is an early and influential result proving that CNNs are well suited to the fine-grained surface-defect recognition task that vehicle damage detection is also an example of. Cracks, scratches and dent detection share a common computer vision challenge: discriminating subtle, often low-contrast, irregular surface features from a visually ‘clean’ background under varying lighting and viewing-angle conditions. Thus, the success of Cha et al. (2017) in the structural-inspection domain is an important piece of supporting evidence that the general approach is viable for the automotive damage domain this dissertation addresses, while also suggesting that lighting and image-acquisition consistency, addressed through the domain-adaptive data collection strategy in Section 4.3, materially affects detection reliability.

2.3 The YOLO Family and Automotive Damage Detection

The original YOLO architecture was presented by Redmon et al. (2016) and framed object detection as a single regression problem, which can be solved in a single forward pass through the network, rather than the multi-stage region-proposal pipelines used in the past. This single-stage design dramatically improves the inference speed, which is directly relevant to a rental damage inspection tool that needs to produce results to staff in near real time during the vehicle check-in/check-out process rather than after a lengthy batch-processing delay.

Later versions of YOLO architecture have gradually closed the gap in accuracy to slower two-stage detectors while maintaining their speed advantage. Mahaur and Mishra (2023) report the mean Average Precision (mAP) for automotive dent and scratch detection using YOLOv7 at 87.3%. This provides direct empirical evidence that the YOLO family can achieve high-accuracy detection for the specific damage classes addressed in this dissertation. This dissertation uses the architecture YOLOv8 as described by Jocher et al. (2023) . YOLOv8 introduces a decoupled detection head and anchor-free design which are reported to improve performance on small-object and multi-class detection tasks in particular, which is directly applicable to this dissertation as scratches and cracks often make up a small proportion of the overall image frame relative to the vehicle body. YOLOv8 was selected and benchmarked against a YOLOv5 baseline as discussed in the proposal, which allows this dissertation to quantify whether the architectural improvements introduced since YOLOv5 provide a measurable accuracy gain on the DriveTime Rentals dataset specifically, as opposed to benchmark results reported on generic datasets.

In addition to architecture selection, the broader object detection literature is also addressing specific failure modes that are relevant to rental inspection photography. Kumar et al. (2021) investigate ensemble detection methods aimed at improving recall when damage is partially occluded, for example by dirt, reflections, or the angle of the photograph, a scenario that is common in non-laboratory, real-world rental-lot photography and therefore directly informs the data augmentation and annotation strategy adopted for this dissertation's dataset. Li et al. (2020) and Zheng et al. (2021) both apply deep learning to the closely adjacent commercial application of insurance-claim vehicle damage assessment. Their results corroborate that deep learning-based damage classification can reach a level of accuracy sufficient for use in a process with real financial consequences, while also noting that classification accuracy degrades for damage types and vehicle finishes that are under-represented in the training set, a limitation this dissertation addresses directly through its domain-adaptive dataset curation.

2.4 Transfer Learning and Domain Adaptation

Training a deep object detector from randomly initialised weights typically requires far more labelled data than is available for a specialised task such as rental vehicle damage detection. Transfer learning, in which a model is first pre-trained on a large, general-purpose dataset such as COCO and then fine-tuned on a smaller, task-specific dataset, is the standard response to this constraint, and is the approach adopted for this dissertation's YOLOv8 model. Nath et al. (2020) provide direct empirical support for this approach, demonstrating that transfer learning is effective for domain-specific classification tasks even when the available labelled dataset is limited in size, a finding of direct relevance given the comparatively modest scale of the dataset being curated for this dissertation relative to the millions of images in COCO.

Domain adaptation, which is the process of adapting a model trained on one data distribution to work well on a related but different data distribution, is closely related to transfer learning and is of particular importance to this dissertation because the publicly available vehicle damage datasets were not collected under rental-lot conditions. The literature on domain shift in computer vision generally indicates that models trained purely on a source domain (here, public datasets) experience a measurable performance drop when deployed on a target domain (here, DriveTime Rentals' rental lot) unless the training data is adapted or supplemented to better represent the target conditions, supporting the dissertation's strategy of supplementing public datasets with real-world DriveTime Rentals images rather than relying on public data alone.

2.5 Datasets for Vehicle Damage Detection

The data curation strategy of the dissertation is based on two public datasets. Wang et al. (2023) present CarDD, a dataset of 4,000 images with pixel-level annotations for six damage categories. At the time of writing, it is one of the few dedicated, peer-reviewed datasets for vehicle damage detection and segmentation. In addition to CarDD, the Roboflow Vehicle Damage Detection dataset is available with about 3,500 more annotated examples, already formatted for YOLO-family models, which decreases the annotation format conversion work required before beginning training.

However, neither dataset was collected under conditions similar to a rental operator's vehicle return bay; the lighting, camera angle, and population of vehicle makes and models all differ from what a deployed system would see in practice. This is the primary data gap that the methodology of this dissertation directly addresses by collecting real-world images from DriveTime Rentals PVT LTD and using domain-adaptive augmentation to create a final training corpus of 6,000-8,000 annotated images that is more representative of true deployment conditions than either public dataset alone.

2.6 Full-Stack Deployment in Computer Vision Enterprise System

A model that works well in a research notebook is not, in and of itself, a usable operational tool. Reddy and Vishnu (2022) consider the practical challenges of deploying computer vision for industrial inspection. They identify the lack of published end-to-end system evaluations, including inference serving, integration with existing business systems, and usability by non-specialist operators as a major barrier to industrial adoption of otherwise mature computer vision techniques. This observation is central to the originality claim of this dissertation: the underlying detection accuracy of YOLO-family models for vehicle damage is reasonably well established (Mahaur and Mishra, 2023; Li et al., 2020), but no published study has evaluated the full pipeline from a Python-based inference microservice through a .NET enterprise backend to an Angular staff-facing dashboard for the specific use case of rental damage inspection.

Sculley et al. (2015) offer a complementary systems-engineering view, warning that machine learning components bring significant 'hidden technical debt' when integrated into a larger software system, such as model and surrounding pipeline entanglement, configuration complexity, and the operational costs of tracking model performance over time. The literature described above informs the architectural decisions described in Section 4.7, including the decision to isolate the YOLOv8 model behind a dedicated FastAPI microservice with a defined contract, rather than embedding the inference logic within the .NET backend, to mitigate the entanglement risk identified by Sculley et al. (2015).

2.7 Explainability, Bias and Ethics in Automated Inspection

Since the output from the system may be used as evidence in a financial dispute between a rental operator and a customer, a binary damage/no-damage classification is not sufficient, the system must give a structured and interpretable record of what was detected, where and with what confidence. Arrieta et al. (2020) provide a comprehensive review of Explainable Artificial Intelligence (XAI), differentiating between model-intrinsic interpretability and post-hoc explanation methods, and contend that the degree of explainability required depends on the stakes and the audience of a given application. For the present dissertation, the practical implication is that annotated bounding boxes with associated class labels and confidence scores exported into a structured PDF report is an appropriate and proportionate form of explainability for a non-technical rental-operations audience, rather than more complex techniques such as saliency mapping.

Bias is another concern and it is directly relevant to a system which will be trained on a finite set of vehicle makes, models and colours drawn primarily from one operator’s fleet. If certain car colours or body types are under-represented in the training data, the resulting model risks exhibiting systematically lower detection accuracy for those vehicles, a fairness concern analogous to those documented in the wider computer vision fairness literature. This concern directly shapes the ethical considerations and dataset evaluation strategy described in Sections 4.8 and 5.

2.8 Conceptual Framework

The conceptual framework of the dissertation synthesizes the above strands, placing the research artefact at the confluence of three previously unconnected bodies of work: (i) deep learning object detection research, which has demonstrated that YOLO-family architectures can achieve high accuracy on automotive damage classes provided sufficient and representative training data (Redmon et al., 2016; Mahaur and Mishra, 2023; Jocher et al., 2023); (ii) software-systems research on deploying machine learning models reliably within larger applications, highlighting the integration and technical-debt risks that must be managed (Sculley et al., 2015; Reddy and Vishnu, 2022); and (iii) responsible-AI research on explainability and bias, which sets requirements for how a system's output must be structured and evaluated before it can be trusted for a use case with real financial consequences (Arrieta et al., 2020). The methodology in Chapter 4 is designed to address all three strands together, rather than treating model accuracy, system integration and responsible deployment as separate, sequential concerns.

2.9 Summer of Key Literature

Table 1 summarizes the most relevant studies to the dissertation design, and relates each study to the relevant gap identified in Chapter 3.

Source	Focus	Key Contribution	Relevance to This Study
Redmon et al. (2016)	Single-stage object detection	Introduced YOLO; real-time detection in one forward pass	Baseline architecture rationale
He et al. (2016)	CNN architecture	Residual learning enabling deeper, more accurate networks	Backbone theory for YOLO
Cha et al. (2017)	Structural defect detection	CNNs match human-level accuracy on crack detection	Validates CV for surface-defect tasks
Wang et al. (2023)	Vehicle damage dataset	CarDD: 4,000 pixel-annotated damage images	Primary public dataset source
Mahaur & Mishra (2023)	YOLOv7 automotive detection	87.3% mAP on dent/scratch detection	Benchmark accuracy target
Jocher et al. (2023)	YOLOv8 architecture	Anchor-free, decoupled head; improved small-object accuracy	Selected model architecture
Nath et al. (2020)	Transfer learning, site safety	Effective fine-tuning on limited labelled data	Justifies transfer-learning strategy
Reddy & Vishnu (2022)	CV deployment in industry	Identifies lack of end-to-end system evaluations	Core systems-integration gap
Sculley et al. (2015)	ML systems engineering	'Hidden technical debt' in ML-embedded systems	Informs microservice architecture
Arrieta et al. (2020)	Explainable AI	Framework for proportionate model explainability	Informs evidentiary report design

Table 1: Summary of key literature underpinning the dissertation's design.

2.10 Research Gap

Chapter 2 reviews literature and it identifies two specific gaps that the current dissertation aims to fill.

The first is a representativeness gap in the data. We used public datasets for vehicle damage detection, such as CarDD (Wang et al., 2023) and the Roboflow Vehicle Damage Detection dataset, which were not collected under the lighting, camera-angle and fleet-composition conditions typical of a car rental return bay. Thus, models trained solely on such public datasets are expected to underperform when deployed in a real rental operational context. To the best of the researcher's knowledge, the domain-shift problem has not been addressed previously for the rental inspection use case specifically. This dissertation fills the gap practically, by supplementing the public datasets, with real-world annotated images collected directly from DriveTime Rentals PVT LTD. The second, and in principle more important, gap is a systems integration gap. Reddy and Vishnu (2022) have shown that the computer vision literature is mainly focused on measuring detection accuracy in isolation from the enterprise systems that would need to consume the detection output in practice. To date, the information systems literature on enterprise software architecture has not examined the specific integration pattern of a Python-based deep learning inference service embedded within a .NET and Angular enterprise stack. To the best of our knowledge, no published study has assessed such a system end-to-end, from raw inspection photograph through to structured, exportable, evidentiary report usable by non-technical rental staff as part of a live operational workflow. This dissertation directly addresses this gap by means of its full-stack design science research artifact and its planned evaluation of model accuracy, API latency, inspection-time reduction and usability as an integrated whole, rather than evaluating the deep learning component alone.

The main novelty of the dissertation is the joint consideration of both gaps resulting in a domain-adapted YOLOv8 detection model. The practical value of this model is demonstrated not only by the benchmark accuracy metrics, but also by its integration and evaluation in a realistic enterprise inspection workflow.

3 METHODOLOGY

3.1 Research Design and Philosophy

The research philosophy underpinning this dissertation is pragmatist and follows the Design Science Research (DSR) paradigm as described by Hevner et al. (2004) and operationalized through the structured process model of Peffers et al. (2007). The dissertation is not a purely theory-testing study, but rather a designed artefact, the YOLOv8 detection model and the surrounding.NET/Angular system, which is evaluated for its utility for solving a real organizational problem. This makes DSR appropriate as the contribution of the dissertation is, at its core, an artefact. DSR directly supports the dual contribution of the dissertation: a validated technical artifact for DriveTime Rentals PVT LTD and generalized design knowledge, in the form of a reusable architectural pattern for integrating Python-based computer vision models into.NET and Angular enterprise systems, that is transferable to other organizations and problem instances.

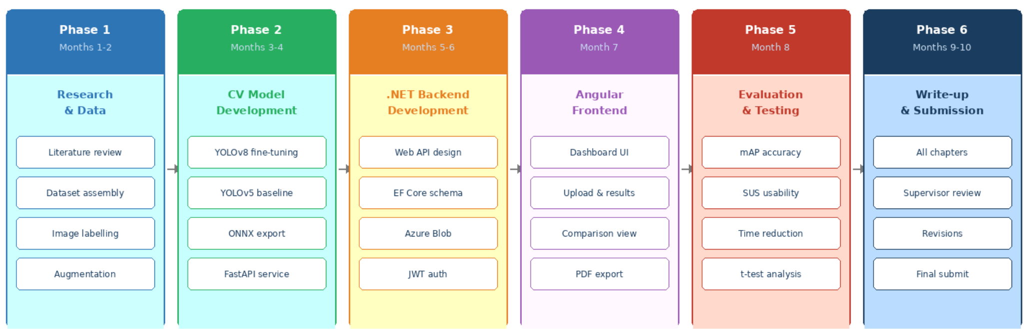

Figure 1: The four-phase Design Science Research process adopted for this dissertation, with current progress marked.

3.2 Research Approach

The study is mixed-methods, i.e. it combines a quantitative assessment of model and system performance with a small-scale qualitative/quantitative usability study. The quantitative strand assesses detection accuracy (mAP@0.5, precision, recall, F1-score), API latency under load, and inspection-time reduction by a paired-sample comparison. The qualitative/quantitative strand employs the System Usability Scale (SUS) questionnaire (Brooke, 1996) to capture staff perceptions of the dashboard, and open feedback obtained at the user study. This mixed approach is consistent with the DSR principle that evaluation of an artefact should be based on its technical performance and practical utility to its intended users (Hevner et al., 2004; Peffers et al., 2007).

3.3 Data Collection Method

The model training dataset is provided by two parallel data collection streams. The first is based on the existing public datasets CarDD (Wang et al., 2023) and the Roboflow Vehicle Damage Detection dataset, providing an initial labelled corpus of about 7.5k images covering the targeted damage classes. The second is primary data collection: real world inspection images collected directly from DriveTime Rentals PVT LTD under a formal data sharing agreement with the operator, capturing the lighting conditions, camera angles and fleet vehicle types the deployed system will actually see. All images from both sources are annotated using the Roboflow annotation platform over five damage classes such as scratch, dent, crack, broken component and no-damage/background. Domain-adaptive augmentation (controlled variation in brightness, rotation and occlusion) is used to increase the combined corpus to a target of 6,000-8,000 annotated images. A secondary, qualitative data stream supporting the usability evaluation includes structured task-completion logs and SUS questionnaire responses gathered from a ten-participant user study.

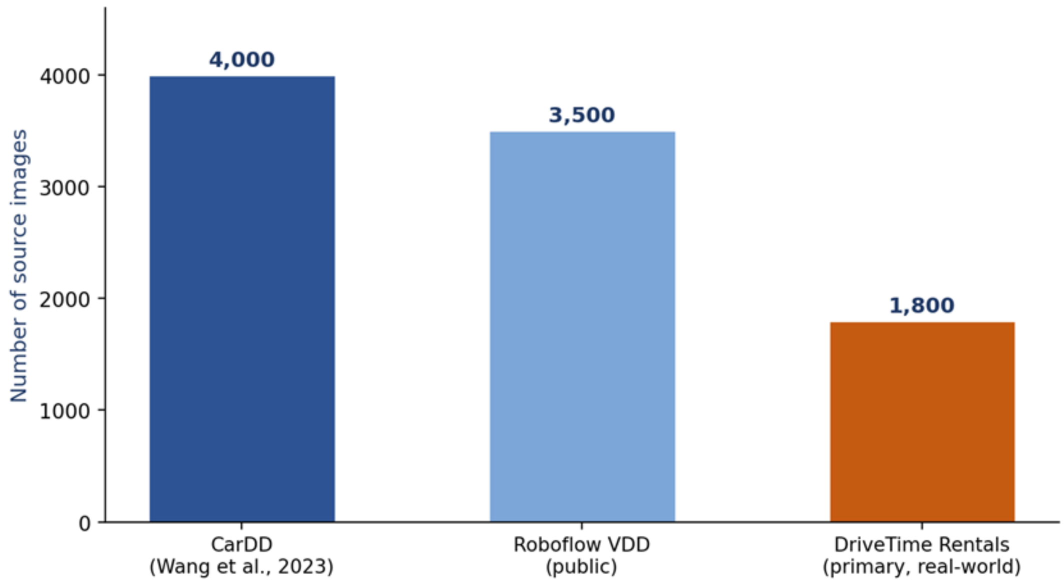

Figure 2: Composition of the combined training dataset by source, prior to augmentation.

3.4 Sampling Technique

Sampling of the image for the DriveTime the rentals section of the dataset uses purposive sampling, where inspection images are deliberately chosen to cover the operator's actual fleet composition (make, model, colour) and the range of lighting conditions experienced in a typical operating day, to maximise representativeness of true deployment conditions rather than maximising raw image count alone. The usability study will use a convenience sample of ten volunteer participants, recruited from the DriveTime Rentals staff and university peers familiar with basic digital tools. The sample size is in line with established guidance for usability testing, where a sample of this size is generally sufficient to surface most significant usability issues. The resulting findings are considered indicative rather than statistically generalisable, as is acknowledged as a limitation in section 1.6.

3.5 Data Analysis Procedure

Model performance will be analysed using standard object detection metrics (mean Average Precision at an IoU threshold of 0.5 (mAP@0.5), precision, recall and F1-score), based on a held-out test set of 800 images not seen during training, with results for the fine-tuned YOLOv8 model directly compared against a YOLOv5 baseline trained on the same dataset split. To analyze system performance, we will load-test the.NET 8 Web API and FastAPI inference microservice using k6 and record the response latency distributions under simulated concurrent usage. The reduction in inspection time will be evaluated with a paired t-test comparing time to complete a structured inspection task manually versus using the prototype system with the same ten participants performing both conditions. The SUS questionnaire usability data will be scored using the standard scoring procedure described by Brooke (1996) and will be compared with the generally accepted value of 70 for acceptable usability. Any open text feedback collected will be thematically analysed.

3.6 Software Development Methodology

The artefact is created with an iterative, agile-inspired approach based on the four DSR phases put forward in the proposal: dataset creation, model development, full-stack implementation and evaluation. In particular, during the full-stack implementation phase, work proceeds in short, incremental development cycles, building and testing one vertical slice of functionality such as image upload through to annotated result display at a time, rather than attempting a single large-bang integration at the end of the project. This incremental approach allows early integration issues between the FastAPI inference service, the .NET 8 Web API and the Angular dashboard to surface and be addressed well before the evaluation phase, aligning with the technical-debt risks identified by Sculley et al. (2015) in Section 2.6.

3.7 Tools and Technologies

The model development stream is implemented in Python 3.11 with the Ultralytics YOLOv8 implementation . Training is performed on Jupyter notebook. Dataset annotation and versioning is handled with Roboflow . The fine-tuned model is exported to ONNX format and exposed behind a Python FastAPI microservice, which is specifically chosen to decouple the inference component from the main application backend, as discussed in Section 2.6. The backend is written in.NET 8, uses Entity Framework Core for database access against a SQL Server database, and stores inspection images in Azure Blob Storage, protected by JWT authenticated access controls. The staff-facing dashboard has been implemented in Angular 17. It provides inspection initiation, multi-image upload, annotated-result display, pre/post comparison and PDF export of inspection reports. All software components are open source or available under the researcher's existing Azure for Students subscription, and academic literature is sourced through the university's IEEE Xplore, ACM Digital Library and Scopus subscriptions.

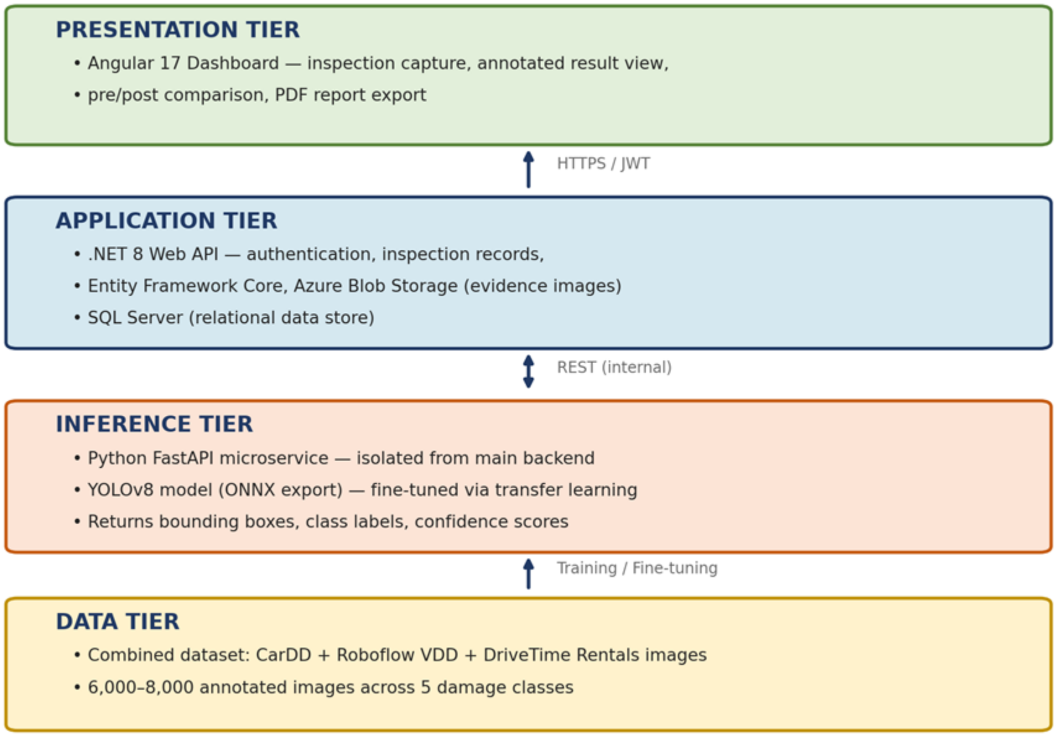

Figure 3: Four-tier system architecture, isolating the inference microservice from the main application backend.

3.8 Ethical Consideration

This research is planned and will be conducted according to the university's research ethics guidelines. All vehicle images on DriveTime Rentals are used under a formal data sharing agreement with the express written permission of the operator and show non-identifiable fleet vehicles only; any personally identifiable information and licence plates visible in source images are blurred before being used in the data set. The combined dataset will be checked for class imbalance over vehicle colour and body type as discussed in Section 2.7 to avoid discriminatory detection performance over different vehicle types. Before participating in the usability study participants sign an informed written consent and may withdraw from the study at any time without penalty. No financial incentives are provided to the participants, thus there is no pressure to participate. The prototype system does not collect or process personal data of rental customers, only vehicle condition and inspection metadata are recorded.

4 INITIAL ARTIFACT PROGRESS

The practical artefact is developed after Phase 1 Dataset Preparation of the methodology presented in the proposal and in Section 4.3. At the point of this Mid-Point Review, there are two parallel streams of activity that are completed or substantially underway: data sourcing and pipeline construction, and an initial data analysis and exploration phase.

4.1 Data Sourcing

We have acquired and examined the class coverage and annotation format of both the public source datasets, CarDD (Wang et al., 2023), and the Roboflow Vehicle Damage Detection (VDD) dataset. CarDD provides COCO format JSON annotations in its original six damage categories (dent, scratch, crack, glass shatter, lamp broken, tire flat) over pre-defined train, validation and test splits. The Roboflow VDD dataset is released as a CSV export with bounding-box coordinates and class labels suitable for YOLO-format training pipelines. Alongside this, real-world inspection photographs were collected from DriveTime Rentals PVT LTD under the agreed data-sharing arrangement, across the operator's fleet at different times of day to provide natural lighting variation, in line with the purposive sampling strategy outlined in Section 4.4.

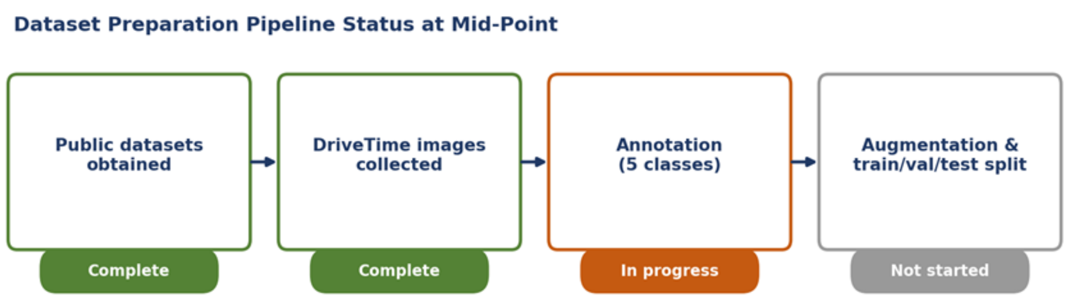

Figure 4: Dataset preparation pipeline status at the Mid-Point Review.

4.2 Data Analysis

The combined dataset is currently being annotated using the Roboflow annotation platform, labelling each image across the five target classes (scratch, dent, crack, broken component, no-damage/background) and is partially complete. To verify the labelling schema, class definitions and bounding-box conventions before scaling the annotation to the full corpus, a working subset of the dataset has been annotated to catch labelling inconsistencies early, rather than after the full dataset has been processed. The domain-adaptive augmentation and the final consolidation of the dataset to the target of 6,000–8,000 annotated images, along with the train/validation/test split (with 800 images reserved for the held-out test set described in Section 4.5), are scheduled to be completed immediately after this submission, ahead of model training in Phase 2.

5 FUTURE PROGRESS

The remaining work is organized into three phases aligned with the supervisory meeting schedule. By late June 2026 (Supervisory Meeting 4), dataset annotation will be finalized and validated, ethical clearance for the usability study will be secured, and the dataset will be prepared for training and evaluation. In July 2026 (Supervisory Meetings 5 and 6), Phase 2 (Model Development) involves fine-tuning the YOLOv8 model using COCO-pretrained weights and establishing a YOLOv5 baseline for comparison. Simultaneously, Phase 3 (Full-Stack Implementation) begins with the development of a .NET 8 Web API and Angular 17 frontend. By August 2026 (Supervisory Meetings 7 and 8), the Angular dashboard will be completed and integrated. Phase 4 (Evaluation) will consist of model testing, usability studies, and statistical analysis. The final dissertation and required documentation are due by 10 September 2026, with risk managed through careful sequencing of high-uncertainty tasks.

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APPENDIX

1) Proportionate Approval Form:
https://drive.google.com/drive/folders/1PwtzwfwTeRxiCq3UoWfPq2R7SrdU5Z68?usp=sharing

2) Log Sheets:
https://drive.google.com/drive/folders/1PwtzwfwTeRxiCq3UoWfPq2R7SrdU5Z68?usp=sharing